## 1.2. Exercises

(1) The inequality $|x-2|<6$ can be expressed in the form $a<x<b$ where $a=$ _____ and $b=$ _____ .

(2) The inequality $-15\le x\le 7$ can be expressed in the form $|x-a|\le b$ where $a=$ _____ and $b=$ _____ .

(3) Solve the equation $|4x+23|=|4x-9|$. Answer: $x=$ _____ .

(4) Find all numbers $x$ which satisfy $|x^2+2|=|x^2-11|$.
Answer: $x=$ _____ and $x=$ _____ .

(5) Solve the inequality $\dfrac{3x}{x^2+2}\ge\dfrac{1}{x-1}$. Express your answer in interval notation.
Answer: [ _____ , _____ ) $\cup$ [ 2 , _____ ) .

(6) Solve the equation $|x-2|^2+3|x-2|-4=0$.
Answer: $x=$ _____ and $x=$ _____ .

(7) The inequality $-4\le x\le 10$ can be expressed in the form $|x-a|\le b$ where $a=$ _____ and $b=$ _____ .

(8) Sketch the graph of the equation $x-2=|y-3|$.

(9) The inequality $|x+4|<7$ can be expressed in the form $a<x<b$ where $a=$ _____ and $b=$ _____ .

(10) Solve the inequality $|3x+7|<5$. Express your answer in interval notation.
Answer: ( _____ , _____ ).

(11) Find all numbers $x$ which satisfy $|x^2-9|=|x^2-5|$.
Answer: $x=$ _____ and $x=$ _____ .

(12) Solve the inequality $\left|\dfrac{2x^2-3}{14}\right|\le\dfrac{1}{2}$. Express your answer in interval notation.
Answer: [ _____ , _____ ].

(13) Solve the inequality $|x-3|\ge 6$. Express your answer in interval notation.
Answer: ( _____ , _____ ] $\cup$ [_____ , _____ ) .

(14) Solve the inequality $\dfrac{x}{x+2}\ge\dfrac{x+3}{x-4}$. Express your answer in interval notation.
Answer: ( _____ , _____ ) $\cup$ [ _____ , _____ ).

(15) In interval notation the solution set for the inequality $\dfrac{x+1}{x-2}\le\dfrac{x+2}{x+3}$ is $(-\infty,$ _____ ) $\cup$ [ _____ , 2 ).

(16) Solve the inequality $\dfrac{4x^2-x+19}{x^3+x^2+4x+4}\ge 1$. Express your answer in interval notation.
Answer: ( _____ , _____ ].

(17) Solve the equation $2|x+3|^2-15|x+3|+7=0$.
Answer: $x=$ _____ , $x=$ _____ , $x=$ _____ , and $x=$ _____ .

(18) Solve the inequality $x\ge 1+\dfrac{2}{x}$. Express your answer in interval notation.
Answer: [ _____ , 0 ) $\cup$ [ _____ , _____ ).